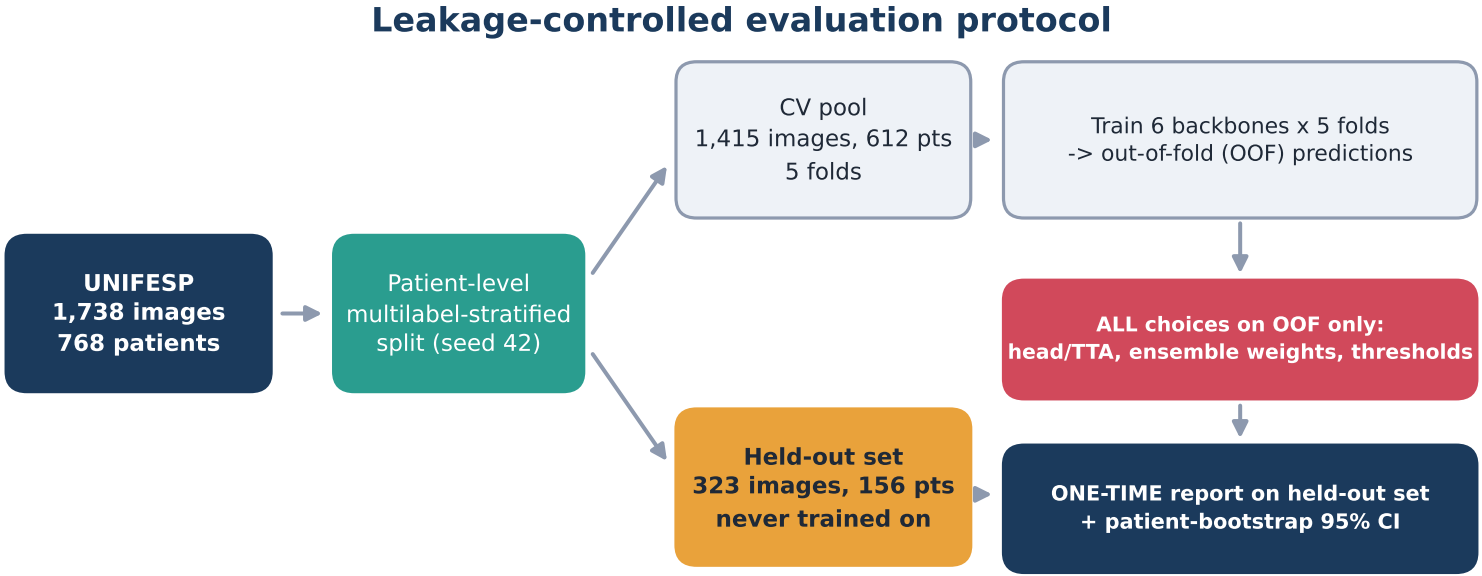
Leakage-controlled evaluation protocol
UNIFESP
1,738 images
768 patients
Patient-level
multilabel-stratified
split (seed 42)
CV pool
1,415 images, 612 pts
5 folds
Train 6 backbones x 5 folds
-> out-of-fold (OOF) predictions
ALL choices on OOF only:
head/TTA, ensemble weights, thresholds
Held-out set
323 images, 156 pts
never trained on
ONE-TIME report on held-out set
+ patient-bootstrap 95% CI
Extensions: ablations - frozen foundation-model probe - MURA / FracAtlas transfer - reliability and label audit - Grad-CAM audit